1. **Aim/Overview of the practical:** Create a Pivot table to quickly summarize large databases, Group within Pivot Tables.

   b) Create multiple Pivot Tables on a single worksheet.

2. **Objective of the practical:** To learn how to create and use Pivot Tables for quick data summarization and grouping in Excel.

   Problem Statement:

   1. Create a PivotTable showing total sales by region
   2. Add a second PivotTable on the same sheet showing average sales by product category
   3. Group sales dates by quarters in your first PivotTable
   4. Add a calculated field to show profit (if your data has cost information)
   5. Insert a slicer to filter both PivotTables simultaneously

   Challenge: Create a third PivotTable that shows the top 5 products by sales volume.

3. **Code for experiment/practical:**

   Step 1:
   First, I created a PivotTable to show Total Sales by Region.
   In the Rows area, placed Region.
   In the Values area, placed Sum of Sales.
   This gave the total sales for each region.

Region	Total Sales by region
East	10146
North	14788
South	5659
West	14810
Grand Total	45403

   Step 2:
   Next, added a Calculated Field to show Profit.
   Sum of Profit was placed in the Values area.
   This allowed comparison of both Sales and Profit together.

Region	Total Sales by region	Sum of Profit
East	10146	3845
North	14788	5791
South	5659	1791
West	14810	5380
Grand Total	45403	16807

Step 3:

Then, grouped the sales data by Quarters in the first PivotTable.

The Date field was grouped into Quarters.

Now the PivotTable showed Region-wise Sales and Profit for each Quarter.

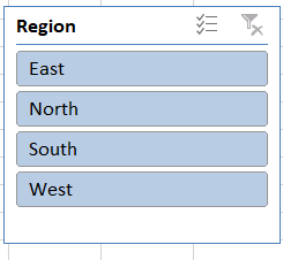
A Column Chart was inserted here to visually display Sales and Profit by Region and Quarter.

Region	Total Sales by region	Sum of Profit
East	10146	3845
2024	10146	3845
Qtr1	2691	889
Qtr2	3297	1275
Qtr3	4158	1681

Step 4:

Inserted a Slicer for Region.

This slicer was connected to both PivotTables so that filtering by Region affected all reports at once.



Step 5:

After that, created a second PivotTable on the same sheet to show Average Sales by Product Category.

In the Rows area, placed Product Category.

In the Values area, placed Average of Sales.

A Bar Chart was created from this PivotTable to show average sales visually.

Product Category	Average Sales
Clothing	1265.416667
Electronics	1120.333333
Furniture	1045
Groceries	882
Toys	1138.2
Grand Total	1135.075

Step 6 (Challenge):

Finally, created a third PivotTable to display the Top 5 Products by Sales Volume.

Product was placed in the Rows area.

Sum of Sales was placed in the Values area.

Products were sorted by Sales in descending order and the top 5 items were selected.

A Pie Chart was inserted to clearly show the proportion of sales among the top 5 products.

Top 5 Products	Sum of Sales
Headphones	3527
Laptop	3750
Car	4312
Doll	4987
Jacket	11711
Grand Total	28287

4. Output:

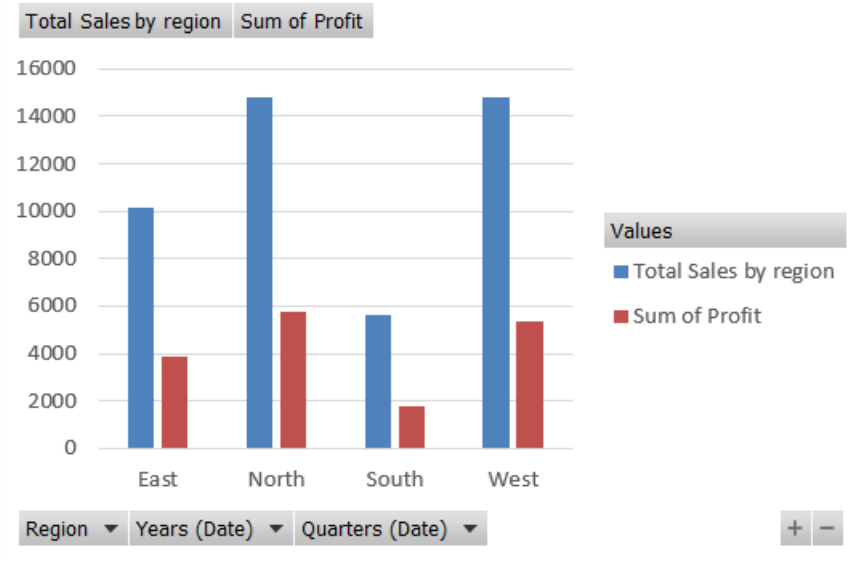
DATASET

Date	Region	Product Category	Product	Sales	Cost
1/1/2024	East	Clothing	Jacket	663	380
1/8/2024	West	Clothing	Jacket	1616	1240
1/15/2024	East	Toys	Doll	165	83
1/22/2024	West	Electronics	Headphon	1332	676
1/29/2024	West	Toys	Doll	551	349
2/5/2024	North	Clothing	Shirt	1529	958
2/12/2024	North	Electronics	Phone	1663	999
2/19/2024	East	Groceries	Milk	835	629
2/26/2024	North	Clothing	Jacket	1040	687
3/4/2024	South	Clothing	Jacket	700	524
3/11/2024	North	Toys	Car	1542	803
3/18/2024	West	Furniture	Chair	1851	1054
3/25/2024	East	Groceries	Bread	1028	710
4/1/2024	North	Electronics	Laptop	827	465
4/8/2024	North	Clothing	Jacket	1400	771
4/15/2024	West	Electronics	Laptop	877	509
4/22/2024	West	Furniture	Chair	569	424
4/29/2024	North	Groceries	Bread	235	132
5/6/2024	North	Electronics	Headphon	1122	694
5/13/2024	South	Electronics	Laptop	385	220
5/20/2024	North	Toys	Doll	1938	1308
5/27/2024	North	Electronics	Phone	1143	740
6/3/2024	East	Clothing	Shirt	1385	758
6/10/2024	South	Toys	Car	888	545
6/17/2024	South	Toys	Doll	1233	934
6/24/2024	East	Clothing	Jacket	1912	1264
7/1/2024	North	Furniture	Chair	701	441

Implementation

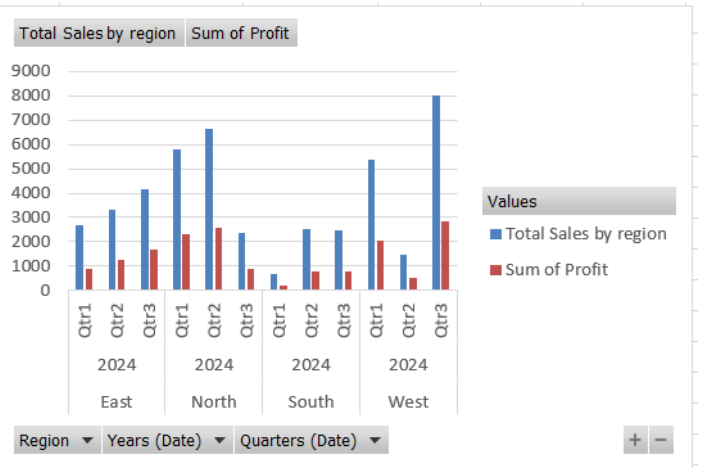
Total sales by region

Region	Total Sales by region
East	10146
North	14788
South	5659
West	14810
Grand Total	45403



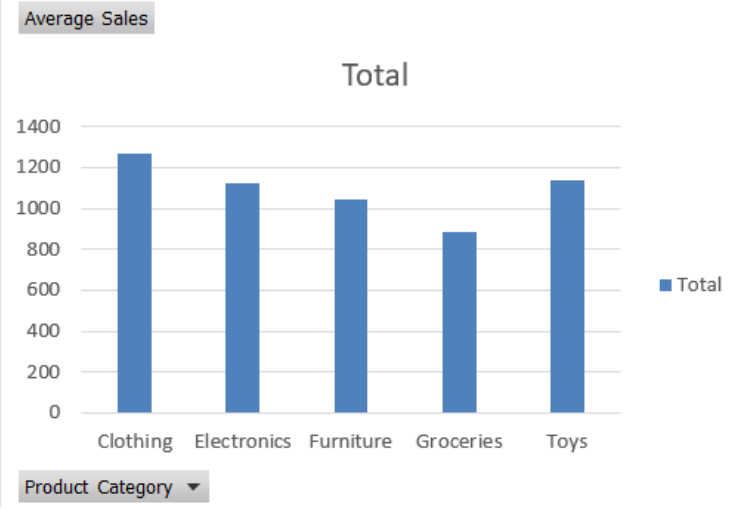
Group sales dates by quarters

Region	Total Sales by region	Sum of Profit
East	10146	3845
2024	10146	3845
Qtr1	2691	889
Qtr2	3297	1275
Qtr3	4158	1681
North	14788	5791
2024	14788	5791
Qtr1	5774	2327
Qtr2	6665	2555
Qtr3	2349	909
South	5659	1791
2024	5659	1791
Qtr1	700	176
Qtr2	2506	807
Qtr3	2453	808
West	14810	5380
2024	14810	5380
Qtr1	5350	2031
Qtr2	1446	513
Qtr3	8014	2836
Grand Total	45403	16807



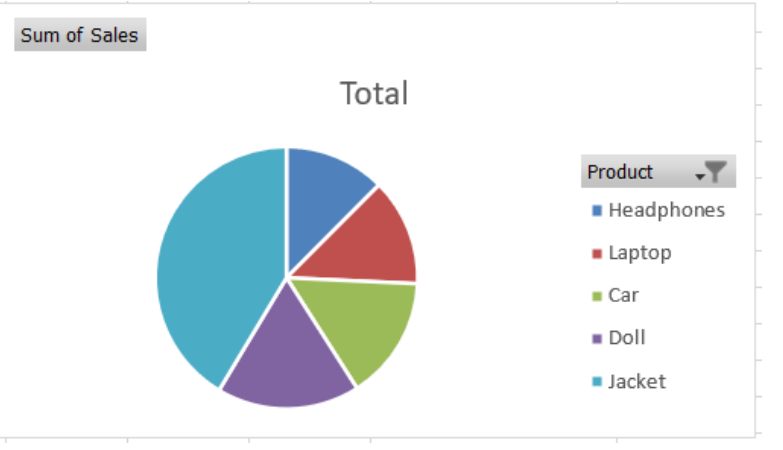
Average sales by product category

Product Category	Average Sales
Clothing	1265.416667
Electronics	1120.333333
Furniture	1045
Groceries	882
Toys	1138.2
Grand Total	**1135.075**

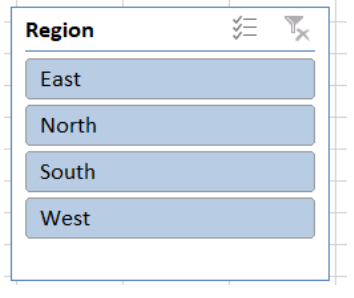


Top 5 products by sales volume.

Top 5 Products	Sum of Sales
Headphones	3527
Laptop	3750
Car	4312
Doll	4987
Jacket	11711
Grand Total	**28287**



Slicer to filter both PivotTables simultaneously



5. Learning Outcomes (What I have learnt):

1. Students will be able to create Pivot Tables to efficiently summarize and analyze large datasets.
2. Students will learn how to group data within Pivot Tables for better insights.
3. Students will be able to create and manage multiple Pivot Tables on a single worksheet for comparative analysis.